

MATHEMATICAL ANALYSIS 1
HOMEWORK 2

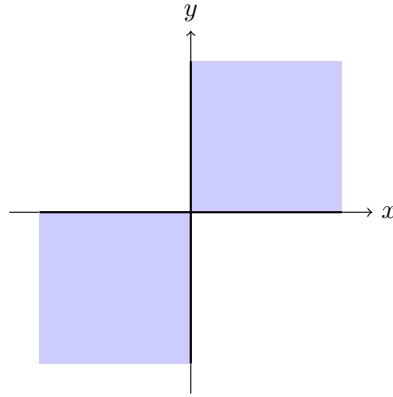
- (1) Describe the following subsets of $\mathbb{R}$, specify what are their infimum and supremum, and determine whether their minimum and maximum are attained (explain your answers).
 - (a) $A = \{x \in \mathbb{R} \mid (x+1)(x-1)(x-5) < 0\} \cap \{x \in \mathbb{R} \mid \frac{3x+1}{x-2} \geq 0\}$
 - (b) $B = \{x \in \mathbb{R} \mid x-4 \geq \sqrt{x^2-6x+5}\} \cup \{x \in \mathbb{R} \mid x+2 > \sqrt{x-1}\}$
 - (c) $C = \{x \in \mathbb{R} \mid x = \frac{1}{n-2}, n = 3, 4, 5, \dots\}$
- (2) Describe and sketch the following subsets of $\mathbb{R}^2$:
 - (a) $A = \{(x, y) \in \mathbb{R}^2 \mid xy \geq 0\}$
 - (b) $B = \{(x, y) \in \mathbb{R}^2 \mid 1 + xy > 0\}$
 - (c) $C = \{(x, y) \in \mathbb{R}^2 \mid x^2 - y^2 > 0\}$
 - (d) $D = \{(x, y) \in \mathbb{R}^2 \mid x - y \neq 0\}$
 - (e) $E = \{(x, y) \in \mathbb{R}^2 \mid |x - y| < 2\}$
 - (f) $F = \{(x, y) \in \mathbb{R}^2 \mid |x - y| < -2\}$
- (3) Compute $\frac{100!}{98!}$. Explain your answer.
- (4) (a) Using the definition $\binom{n}{k} = \frac{n!}{k!(n-k)!}$, prove Pascal's rule: for all integers n and k with $1 \leq k \leq n-1$,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$
 (b) Prove Newton's binomial formula. *Hint: prove by induction, using part (a).*
- (5) For each of the following sequences, determine whether there exists N such that the sequence is (monotonically) increasing or decreasing on $\{N, N+1, \dots\}$. If so, is it strictly increasing/decreasing there? Find the smallest such N . (Explain your answers.)
 - (a) $a_n = \frac{n}{n+1}, n \in \mathbb{N}$
 - (b) $a_n = \frac{(-1)^n}{n}, n \in \mathbb{N}_+$
 - (c) $a_n = n^2 - 5n, n \in \mathbb{N}$
 - (d) $a_n = \frac{2^n}{n!}, n \in \mathbb{N}$
 - (e) The Fibonacci sequence, defined recursively by $a_0 = a_1 = 1$ and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$. Proceed as follows:
 - (i) Prove by induction that for every $n \in \mathbb{N}$, both $a_n > 0$ and $a_{n+1} > 0$. (*Hint: take as the predicate $p(n) = 'a_n > 0 \text{ and } a_{n+1} > 0'$.*)
 - (ii) Show that $a_{n+1} - a_n = a_{n-1}$ for all $n \geq 1$, and use (i) to answer the question.
- (6) For the sequence $a_n = (-1)^n \frac{n}{n+1}, n \in \mathbb{N}$, find a subsequence that is increasing and a subsequence that is decreasing.
- (7) Describe the following sets: (explain your answers)
 - (a) $\text{dom}(f)$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = \frac{1}{\sin x}$.
 - (b) $\text{im}(f)$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = \frac{1}{\sin x}$.
 - (c) $\text{dom}(f)$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = \frac{1}{2+\sin x}$.
 - (d) $\text{dom}(f)$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = 10^x$.
 - (e) $\text{im}(f)$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = 10^x$.
 - (f) $\text{dom}(f)$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = \log_{10}(x)$.
 - (g) $\text{im}(f)$ where $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = \log_{10}(x)$.
- (8) For the following functions f and subsets $B \subseteq \mathbb{R}$, describe $f^{-1}(B)$.
 - (a) $f: \{3, 4, 5, \dots\} \rightarrow \mathbb{R}$ defined by $f(n) = \frac{1}{n-2}$, and $B = (0, 1)$.

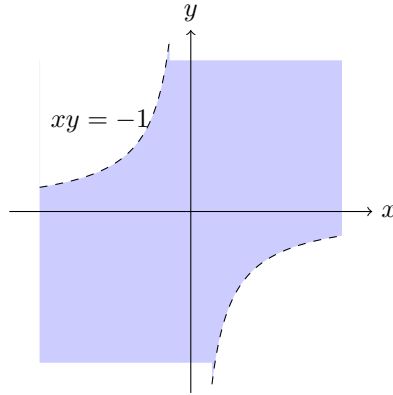
- (b) In the previous question, what if $B = [0, 1]$?
 - (c) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^4$, $B = [1, 16]$.
 - (d) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3 + 1$, $B = (-26, -7]$.
 - (e) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \cos x$, $B = \{0\}$.
- (9) Let $f : X \rightarrow Y$ be a function between two sets X and Y .
- (a) Prove that $A \subseteq f^{-1}(f(A))$ for any $A \subseteq X$.
 - (b) Give an example of when $A \neq f^{-1}(f(A))$.

SOLUTIONS

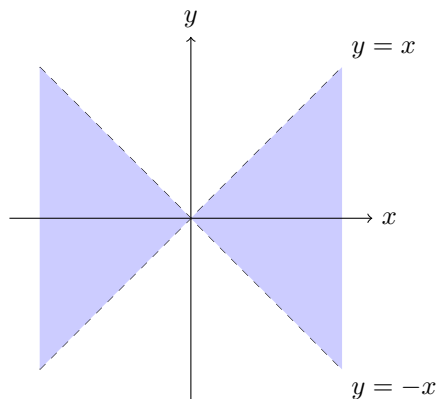
- (1) (a) $(x+1)(x-1)(x-5) < 0$ gives $x \in (-\infty, -1) \cup (1, 5)$. For $\frac{3x+1}{x-2} \geq 0$, we need $(3x+1)(x-2) \geq 0$ and $x \neq 2$, giving $x \in (-\infty, -\frac{1}{3}] \cup (2, \infty)$. Thus $A = (-\infty, -1) \cup (2, 5)$. We have $\inf A = -\infty$, $\sup A = 5 \notin A$. The minimum is not attained, the maximum is not attained.
- (b) For $x-4 \geq \sqrt{x^2-6x+5}$, we first need the square root to be defined, i.e. $x^2-6x+5 = (x-1)(x-5) \geq 0$ (so $x \leq 1$ or $x \geq 5$). Moreover, since a square root is always non-negative, we need the left-hand side to be non-negative as well, i.e. $x \geq 4$. Together these give $x \geq 5$. For such x both sides are non-negative, so squaring gives an equivalent inequality: $(x-4)^2 \geq x^2-6x+5$, i.e. $-8x+16 \geq -6x+5$, i.e. $x \leq \frac{11}{2}$. So the first set is $[5, \frac{11}{2}]$. (Without the condition $x \geq 4$, squaring would wrongly admit all $x \leq 1$.) For $x+2 > \sqrt{x-1}$, we need $x \geq 1$; then $x+2 \geq 3 > 0$, so we may square: $(x+2)^2 > x-1$, giving $x^2+3x+5 > 0$, which holds for all $x \geq 1$. Thus $B = [1, \infty)$, with $\inf B = \min B = 1$, $\sup B = +\infty$.
- (c) $C = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$. We have $\inf C = 0 \notin C$, $\sup C = \max C = 1 \in C$.
- (2) (a) A consists of the first and third quadrants (including axes).



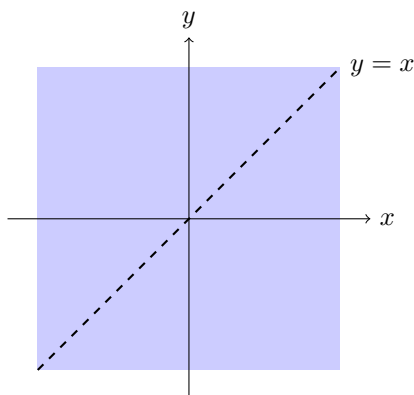
- (b) B is the region between the two branches of the hyperbola $xy = -1$ (excluding the hyperbola itself)



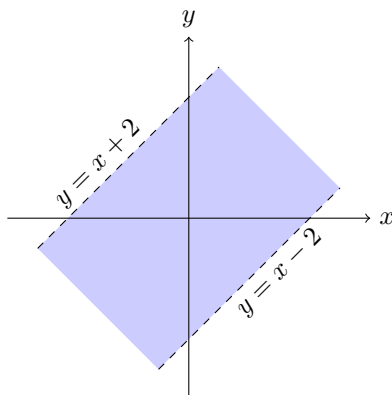
- (c) C consists of regions where $|x| > |y|$, i.e., the regions to the left and right of the lines $y = \pm x$ (excluding the lines themselves).



(d) $D = \mathbb{R}^2 \setminus \{(x, y) \in \mathbb{R}^2 \mid y = x\}$ (i.e., the plane $\mathbb{R}^2$ without the line $y = x$).



(e) E is the strip between the parallel lines $y = x - 2$ and $y = x + 2$ (excluding these lines).



(f) $F = \emptyset$ since $|x - y| \geq 0$ cannot be < -2 .

(3) $\frac{100!}{98!} = \frac{100 \cdot 99 \cdot 98!}{98!} = 100 \cdot 99 = 9900$.

(4) (a) Bringing both fractions to the common denominator $k!(n-k)!$ (using $k! = k \cdot (k-1)!$ and $(n-k)! = (n-k) \cdot (n-1-k)!$):

$$\begin{aligned} \binom{n-1}{k-1} + \binom{n-1}{k} &= \frac{(n-1)!}{(k-1)!(n-k)!} + \frac{(n-1)!}{k!(n-1-k)!} = \frac{(n-1)!k}{k!(n-k)!} + \frac{(n-1)!(n-k)}{k!(n-k)!} \\ &= \frac{(n-1)!(k + (n-k))}{k!(n-k)!} = \frac{n!}{k!(n-k)!} = \binom{n}{k}. \end{aligned}$$

- (b) We prove by induction on n that $(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$. For $n=0$, we have $(a+b)^0 = 1 = \binom{0}{0} a^0 b^0$. Assume the formula holds for $n-1$. Then

$$\begin{aligned}
 (a+b)^n &= (a+b)(a+b)^{n-1} = (a+b) \sum_{k=0}^{n-1} \binom{n-1}{k} a^k b^{n-1-k} \\
 &= \sum_{k=0}^{n-1} \binom{n-1}{k} a^{k+1} b^{n-1-k} + \sum_{k=0}^{n-1} \binom{n-1}{k} a^k b^{n-k} \\
 &= \sum_{k=1}^n \binom{n-1}{k-1} a^k b^{n-k} + \sum_{k=0}^{n-1} \binom{n-1}{k} a^k b^{n-k} \\
 &= a^n + \sum_{k=1}^{n-1} \left(\binom{n-1}{k-1} + \binom{n-1}{k} \right) a^k b^{n-k} + b^n \\
 &= \sum_{k=0}^n \binom{n}{k} a^k b^{n-k},
 \end{aligned}$$

where we used part (a) for $1 \leq k \leq n-1$. (In the lecture notes the formula is written as $\sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$; this is the same statement with the roles of a and b exchanged, or equivalently follows from $\binom{n}{k} = \binom{n}{n-k}$.)

- (5) (a) We have $a_{n+1} - a_n = \frac{n+1}{n+2} - \frac{n}{n+1} = \frac{(n+1)^2 - n(n+2)}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)} > 0$ for all $n \in \mathbb{N}$. So the sequence is strictly increasing on all of $\mathbb{N}$, i.e. $N=0$.
- (b) The terms alternate in sign: $a_n < 0$ for n odd and $a_n > 0$ for n even. Hence for every n odd we have $a_n < a_{n+1}$, and for every n even we have $a_n > a_{n+1}$. So on any set $\{N, N+1, \dots\}$ the sequence is neither increasing nor decreasing, and no such N exists.
- (c) We have $a_{n+1} - a_n = (n+1)^2 - 5(n+1) - n^2 + 5n = 2n - 4$. This is < 0 for $n \in \{0, 1\}$, $= 0$ for $n = 2$, and > 0 for $n \geq 3$. (Indeed $a_0 = 0$, $a_1 = -4$, $a_2 = a_3 = -6$, $a_4 = -4$.) Hence the sequence is increasing on $\{2, 3, \dots\}$ (smallest $N = 2$) and strictly increasing on $\{3, 4, \dots\}$ (smallest $N = 3$). It is not decreasing on any $\{N, N+1, \dots\}$.
- (d) All terms are positive, and $\frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} = \frac{2}{n+1}$. This is > 1 for $n = 0$, $= 1$ for $n = 1$ and < 1 for $n \geq 2$. (Indeed $a_0 = 1$, $a_1 = a_2 = 2$, $a_3 = \frac{4}{3}$.) Since $a_n > 0$, $a_{n+1} \leq a_n$ is equivalent to $\frac{a_{n+1}}{a_n} \leq 1$. Hence the sequence is decreasing on $\{1, 2, \dots\}$ (smallest $N = 1$) and strictly decreasing on $\{2, 3, \dots\}$ (smallest $N = 2$). It is not increasing on any $\{N, N+1, \dots\}$.
- (e) (i) Let $p(n) = 'a_n > 0 \text{ and } a_{n+1} > 0'$. *Base case:* $a_0 = a_1 = 1 > 0$, so $p(0)$ is true. *Induction step:* assume $p(n)$, i.e. $a_n > 0$ and $a_{n+1} > 0$. Then $a_{n+2} = a_{n+1} + a_n > 0$, so $a_{n+1} > 0$ and $a_{n+2} > 0$, i.e. $p(n+1)$ is true. By induction, $p(n)$ holds for all $n \in \mathbb{N}$; in particular $a_n > 0$ for all n . (Each term depends on the two previous ones, which is why the predicate involves two consecutive terms.)
- (ii) For $n \geq 1$, the recursion with $n+1$ in place of n gives $a_{n+1} = a_n + a_{n-1}$, i.e. $a_{n+1} - a_n = a_{n-1}$. By (i), $a_{n-1} > 0$, so $a_{n+1} > a_n$ for all $n \geq 1$: the sequence is strictly increasing on $\{1, 2, \dots\}$. Since $a_0 = a_1$, it is increasing on all of $\mathbb{N}$ (smallest $N = 0$), but strictly increasing only from $N = 1$ on.

Remark. In (c), (d) and (e) the smallest N from which the sequence is monotone differs from the smallest N from which it is *strictly* monotone, because two consecutive terms are equal ($a_2 = a_3$ in (c), $a_1 = a_2$ in (d), $a_0 = a_1$ in (e)). The question does not specify which one is meant, so either answer is acceptable, provided it is justified and it is clear whether it refers to monotone or strictly monotone.

- (6) The subsequence of even indices, $n_k = 2k$, gives $a_{2k} = \frac{2k}{2k+1} = 1 - \frac{1}{2k+1}$, which is strictly increasing in k , since $\frac{1}{2k+1}$ decreases as k increases. The subsequence of odd indices, $n_k = 2k+1$, gives $a_{2k+1} = -\frac{2k+1}{2k+2} = -1 + \frac{1}{2k+2}$, which is strictly decreasing in k , since $\frac{1}{2k+2}$ decreases as k increases.
- (7) (a) $\text{dom}(f) = \mathbb{R} \setminus \{k\pi \mid k \in \mathbb{Z}\}$ (where $\sin x \neq 0$).

- (b) $\text{im}(f) = (-\infty, -1] \cup [1, \infty)$ (since $|\sin x| \leq 1$ with equality attained).
 - (c) $\text{dom}(f) = \mathbb{R}$ (since $2 + \sin x \geq 1 > 0$ for all x).
 - (d) $\text{dom}(f) = \mathbb{R}$, since 10^x is defined for every real exponent x .
 - (e) $\text{im}(f) = (0, \infty)$: we have $10^x > 0$ for all x , and every $y > 0$ is attained, namely $y = 10^{\log_{10} y}$.
 - (f) $\text{dom}(f) = (0, \infty)$, since $\log_{10} x$ is defined exactly for $x > 0$.
 - (g) $\text{im}(f) = \mathbb{R}$: every $y \in \mathbb{R}$ is attained, namely $y = \log_{10}(10^y)$ with $10^y \in (0, \infty)$.
- (8) (a) We need $\frac{1}{n-2} \in (0, 1)$, so $\frac{1}{n-2} > 0$ and $\frac{1}{n-2} < 1$. This gives $n > 2$ and $n > 3$, so $n \geq 4$. Thus $f^{-1}(B) = \{4, 5, 6, \dots\}$.
- (b) $f^{-1}([0, 1]) = \{3, 4, 5, \dots\}$ (including $n = 3$ where $f(3) = 1$).
- (c) $x^4 \in [1, 16)$ gives $|x| \in [1, 2)$, so $f^{-1}(B) = (-2, -1] \cup [1, 2)$.
- (d) $x^3 + 1 \in (-26, -7]$ gives $x^3 \in (-27, -8]$, so $x \in (-3, -2]$. Thus $f^{-1}(B) = (-3, -2]$.
- (e) $\cos x = 0$ when $x = \frac{\pi}{2} + k\pi$ for $k \in \mathbb{Z}$. Thus $f^{-1}(\{0\}) = \{\frac{\pi}{2} + k\pi \mid k \in \mathbb{Z}\}$.
- (9) (a) Let $x \in A$. Then $f(x) \in f(A)$, so $x \in f^{-1}(f(A))$.
- (b) Let $X = \{1, 2\}$, $Y = \{a\}$, $f(1) = f(2) = a$, and $A = \{1\}$. Then $f(A) = \{a\}$ and $f^{-1}(f(A)) = f^{-1}(\{a\}) = \{1, 2\} \neq A$.